# Classic hedging is all you need

Classical versus learned option hedging under transaction costs

*Sepp (2013) and Buehler et al. (2019) replications, plus a fair head-to-head*

DSC4107 Final Project

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# Expectation vs fair test

- Deep hedging is sold as freer → should beat classical bands once costs bite.
- Sepp (2013): closed-form Sharpe-optimal delta bands. Buehler et al. (2019): network on CVaR of terminal P&L.
- Neither paper runs both on one simulator with a classical baseline **re-tuned per cell**.

**We run that comparison — shared rollout, four models, five costs, disjoint seeds, frictionless delta recovery as the correctness anchor.**

# Fair test (light)

**One testbed**

- GBM, jump-diffusion, Heston, Heston+ jumps — Sepp parameters
- ATM one-month call; proportional cost on every trade
- Classical side re-tuned in every map cell

**Controls**

- Learned MLP: moneyness, time, position; CVaR $\alpha = 0.65$
- Disjoint train / val / tuning / search / holdout pools
- Three torch seeds per reported mean

Frozen benchmark: Heston at 20 bp with 25% implied vs 20% realised vol (volatility risk premium).

# 20 bp: the headline

- Learned − classical Sharpe edges **−0.09 to +0.01** — essentially flat
- Jump models: classical slightly ahead
- Holdout on 5 seeds has mean edge **+0.0339**. The edge is real but tiny, and the tail is worse on every seed.

**Classic hedging is all you need at nominal friction. Deep is not a uniform upgrade.**

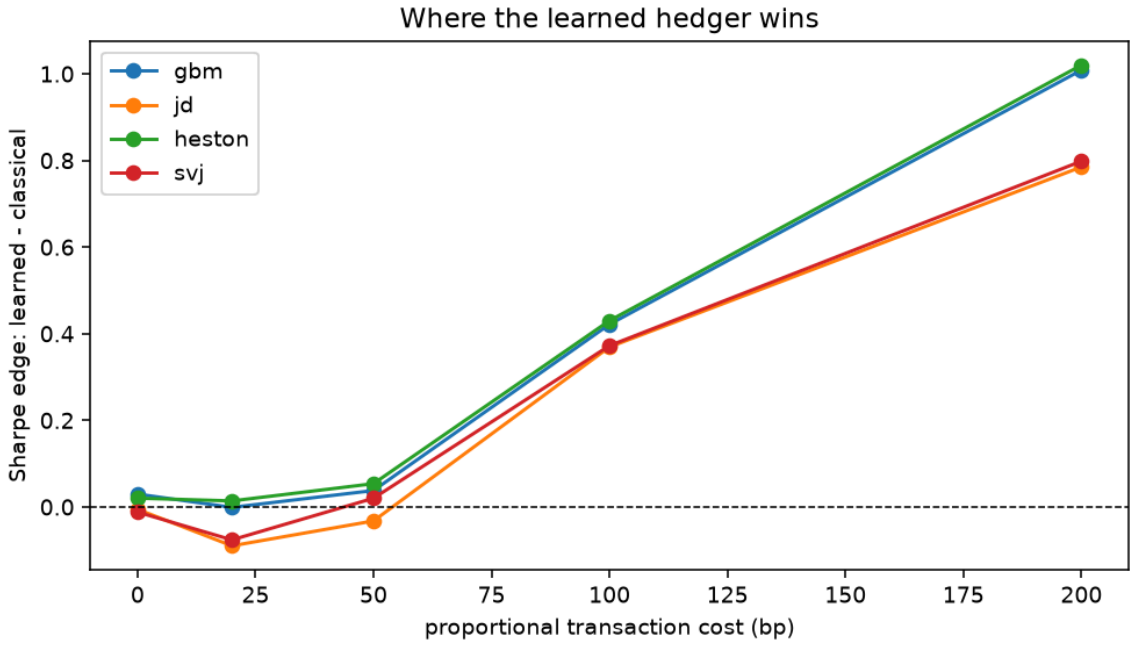



Figure 1: Learned minus classical Sharpe (fixed hyperparameters).

# Replication checks (credibility)

**Sepp**

- Table 2: largest $u$ error 3.9%; case I GBM/SVJ to 0.0001
- JD fix: restore $q$; analytic Sharpe 0.65 matches MC
- 60,000-path cross-check on mean and vol

**Deep hedging**

- Zero cost: corr with BS delta 0.9907
- Heston qual. claim reproduced — beats every-step delta as cost rises

Paper table is not the headline; the 20 bp map is.

# Same spread, worse tail — and the high-cost twist

- Benchmark cell: mean 0.240 vs 0.216; std 0.626 vs 0.613
- Skew −0.70 vs −0.01; CVaR 95% **24% worse**
- **Above 50 bp:** Sharpe "wins" with std → 3–4, mean → premium
- Paired spread and tail — **premium harvesting**, not better hedges

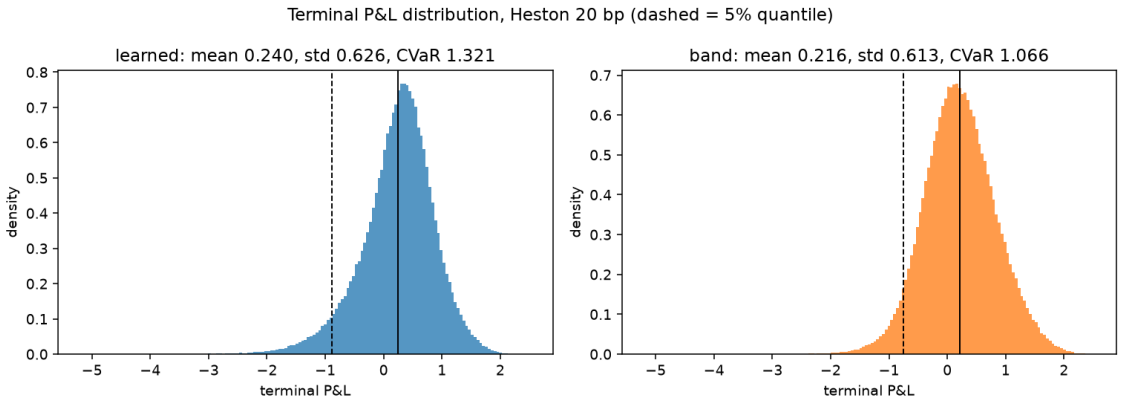



Figure 2: Terminal P&L on the frozen benchmark.

# Liquid 10 bp (footnote only)

- Sepp Case I: **k/2** on $|\Delta\delta| \cdot S$ (eq. 2.5); side study at liquid-index **k** uses our **full-rate** ledger (`cost = 0.001` — numerically Sepp's **k**, not Case I economics)
- 10 bp classical hull re-swept; not `classical_best.json`
- H **clears** (narrow) → K **dominated** → L **a05_only** → I **fails** → J **band_like**

Recipe-specific, mandate-dependent, brittle. Does not reopen the 20 bp headline.

# Conclusion

- At 20 bp a re-tuned classical band does not lose on Sharpe; any tiny learned edge trades tail for mean.
- High-cost Sharpe gains track under-hedging — report them with P&L std and CVaR.
- Replications hold: Sepp Table 2, Monte Carlo cross-check, frictionless delta recovery.

github.com/hluaguo/dsc4107

Full tables: `submission/report/report.pdf`.